

Scalability & Future Plan

Date	06 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	1Mark

Scalability & Future Plan:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Single Machine Learning Model	Accuracy may vary for different user segments and unseen data.	High
2	Limited Dataset Size	Model performance can be improved with more diverse and larger datasets.	Medium
3	Basic Web Interface	User experience and UI/UX can be enhanced for better usability.	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Current application handles limited users on a single Render instance.	Deploy on scalable cloud platform with auto-scaling (AWS / Azure / GCP) and load balancing to support multiple concurrent users.
2	Data Storage	Currently using local files / CSV for data storage.	Migrate to a cloud database (PostgreSQL / MySQL / MongoDB) for secure, reliable and scalable data storage.
3	Performance	Model prediction time may increase with large data and high user load.	Optimize the ML model, use efficient data preprocessing, model compression and caching to reduce prediction time and improve response.
4	Security	Basic security with form validation and HTTPS.	Implement user authentication, input sanitization, rate limiting and encrypted data storage for better security.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Add user authentication (Register / Login) and maintain user history.	Short Term (0 – 3 months)	Improves data security and provides personalized experience to users.
Phase 2	Integrate advanced ML models and retrain with larger, updated datasets.	Medium Term (3 – 6 months)	Improves prediction accuracy and model reliability.
Phase 3	Add new features like Credit Score Analysis, Loan Amount Recommendation and EMI Calculator.	Medium Term (6 – 9 months)	Enhances functionality and helps users make better financial decisions.
Phase 4	Develop mobile app / PWA and API integration for third-party services.	Long Term (9 – 12 months)	Expands accessibility, improves user engagement and supports future growth.